

Reflex — Event-Driven Delivery Management

Small Kenyan Retailers - Real-Time Delivery
Tracking

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Date: September 5, 2026

Small Retailers Manage Deliveries via WhatsApp — No Tracking

- . Retailers coordinate deliveries over WhatsApp and phone calls
- . No record of who's assigned to which delivery
- . No status visibility – "Where is the order?"
- . No proof of delivery
- . Result: Lost orders, frustrated customers, wasted time

E.g., "A retailer loses an order because they forgot which rider took it."

Reflex – One System for All Delivery Tracking

- . Retailer logs a delivery request
- . Dispatcher assigns it to a rider
- . Rider updates the status in real-time
- . Everyone knows where every delivery stands

Transparency · Accountability · Real-Time Visibility

Event-Driven Architecture: Queue + Webhooks + SSE

Retailer — (Requests)

Queue — (Pending)

Webhook — (Completed)

UI Shows — "Pending"

Assigns — Dispatcher

Delivers — Rider

Queue — Webhook — SSE = Real-Time, Event-Driven System

Delivery Status: Requested - Assigned - Picked Up - Delivered

Delivery {

id: string (e.g., "DEL-A1B2C3")

customer_name: string

status: "Requested" | "Assigned" | "Picked Up" | "Delivered"

assigned_rider: string | null

delivered_at: timestamp

}

Every status change goes through: Queue - Worker - Webhook - Real-time update

1. Retailer logs a delivery - Publishes to queue - UI shows "Requested"

2. Dispatcher assigns to a rider - Webhook confirms - UI shows "Assigned"

3. Rider marks "Picked Up" - Webhook confirms - UI shows "Picked Up"

4. Rider marks "Delivered" - Webhook confirms -s
UI shows "Delivered" + Proof

Retailer, Dispatcher, Rider – Each Has a Role

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Retailer	Dispatcher	Rider
Logs delivery requests	Sees open requests	Sees assigned deliveries
Tracks status in real-time	Assigns to a rider	Updates status(picked up, delivered)

Made Conscious Trade-offs — Here's Why

1. In-Memory Storage vs. Database

- Accepted because: Prototype, need to ship fast
- Cost: Data lost on restart
- Future: Swap with SQLite

2. SSE vs. WebSockets

- Accepted because: Simpler, server–client only
- Cost: No bi-directional communication
- Future: Switch if needed

3. In-Memory Queue vs. RabbitMQ

- Accepted because: No external dependencies
- Cost: Queue not persistent
- Future: Use Redis for production

1. Duplicate Assignments

- Second assignment rejected with 409 Conflict
- 'assigned_deliveries' set prevents duplicated

2. Out-of-Order Webhooks

- Status transitions validated (can't skip Picked Up)
- Duplicate protection still holds

3. Worker Thread Resilience

- Per-job thread survives Gunicorn process forking
- Works on Render free tier

What's Next for Reflex

1. Persistent Storage

- SQLite to survive server restarts

2. Retry Logic

- Exponential backoff for failed webhooks

3. QR Scanner Integration

- Real QR code scanning via webcam

4. Admin Dashboard

- Real-time metrics and analytics

5. Containerization

- Docker + docker-compose for easy deployment